Recorded processing covariates map a track onto its counterfactual on trained pairs, but the map does not carry to unseen pairs

Mehdi Foroozandeh

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Abstract

We asked whether recorded processing covariates carry enough information to map one measurement of an epigenomic track onto another of the same material. We built counterfactual arms of 7 tracks, each changing one of 10 processing settings, and trained a generator that reads the covariates and outputs the map. On trained depth pairs, the better designs close 0.77 to 0.92 of the gap from a covariate-free monotone map to the exact oracle (seed range at most 0.53 of that gap). On unseen pairs the gain over a scrambled-covariate twin falls from 0.29 to 0.11, and no design meets the depth law. All readings are exploratory, not judged.

Keywords: epigenomics, experimental covariates, counterfactual data, sequencing depth

1 Motivation and background

CANDI imputes epigenomic tracks from raw counts and reads experimental covariates (depth, run type, read length) on both sides; its zero-shot claims rest on that conditioning. Inside CANDI it cannot be tested alone: training tracks differ in biology and covariates at once, and the covariates are collinear (they change together). Depth has a closed form, binomial thinning; the other settings have none. Missing was data where one setting's true effect is known. Our question: do the recorded covariates carry enough information to map one measurement of a track onto another of the same material?

2 Methods

Counterfactual arms.

We took 7 ENCODE imputation-challenge tracks (one DNase-seq, three narrow and three broad histone marks), each processed to a base of 30M reads (DNase 50M) and to arms that change one setting: depth (15M, 7.5M, 3.75M subsamples of the base), read length (36, 50 bp), mapping filter, duplicate removal, paired-end, an antibody proxy (50% or 90% control reads), fragment extension and three control settings: 130 products. Each product was split into two pseudoreplicate halves (random halves of its reads).

Model and competitors.

A generator g reads the source and target covariates C, C' and outputs a map f; $f(X)$ predicts the target as negative binomial (NB) counts per 25 bp bin, or as $-\log_{10} p$ (Fig. 1). Four designs

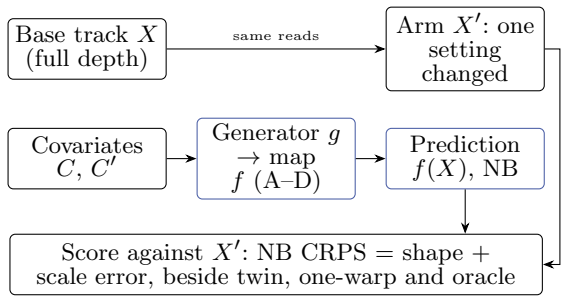


Fig. 1 Design. Each arm reprocesses the base's reads with one setting changed. The generator reads both covariate sets and outputs the map that turns the base into a predicted arm, scored beside three competitors.

of f rise in capacity: per-bin affine (A), per-bin monotone curve (B), kernel plus curve (C) and a conditioned convolutional network (D). The competitors are the no-covariates twin (covariates scrambled), one-warp (a covariate-free monotone map per pair) and an oracle. The score is the NB CRPS (continuous ranked probability score, distance from forecast to observation), split into a shape part and a scale error. Gap closed is $(D_{\text{one-warp}} - D_{\text{model}})/(D_{\text{one-warp}} - D_{\text{oracle}})$. The bar, set before reading, is gap closed ≥ 0.5 in every mark class and a gain above $2\times$ the seed range (largest minus smallest of 3 seeds). Scoring uses held-out chr19 and chr21.

Floor and maps.

The resampling floor thins the halves along a depth ladder and reads 1 kb shape (one minus Spearman) and peak height. Regional maps fit signal level as the null, then read GC, mappability, repeats and peak context, beside a shuffle control.

Two corrections.

First, the floor was planned as the depth oracle. But a depth arm shares its base's reads, so the floor sat above one-warp and inverted the interval. The oracle is now the exact nested-read oracle: binomial thinning of the base count, and its Bayes posterior in reverse. The real arms have 0.994 to 1.005 of the binomial variance. Second, a read adds one count to each 25 bp bin it covers ($k = 4$–5); the first floor thinned single counts, so multi-bin sums were about $\sqrt{k}$ too smooth. We rebuilt it by thinning whole reads.

3 Results and discussion

Depth: the covariates close most of the gap.

From base to arm, designs B to D close 0.77 to 0.92 of the gap in every mark class (Fig. 2a), with a seed range of 0.02 to 0.53 in gap units; the gain beats $2\times$ the seed range in 18 of 18 cells. In narrow marks, design B scores 0.175 (0.174 shape + 0.0008 scale error), one-warp 0.197 (0.192 + 0.0057) and the oracle 0.169 (0.169 + 0.0000), against $2\times$ seed range 0.0015. The interval is small (0.028 to 0.031). Design A shows no gain in DNase and narrow marks (-0.62 to -0.01, seed range 0.7 to 2.6). From arm to base, C and D pass the per-bin oracle (1.11 to 3.68) by reading neighbouring bins.

Run type: better than one-warp, no valid ceiling.

Paired-end reprocessing is not a subsample, so the floor stays its oracle. In narrow marks the floor, 1.004 (0.500 + 0.058), lies above one-warp, 0.815 (0.776 + 0.038), so gap closed cannot be read. The model beats one-warp by 0.055 to 0.213 in 16 of 16 histone cells, each above $2\times$ the seed range (0.002 to 0.065); in DNase, 1 of 8 cells.

Unseen pairs.

The relative count gain over the twin falls from 0.29 on trained pairs to 0.11 on unseen pairs of one setting (0.16 for two); in p, from 0.15 to 0.06 (0.004). It clears $2\times$ the seed range in 14 of 16 configurations trained and 7 of 16 unseen. Only the control settings keep their p gain (0.36 to 0.62 trained, 0.31 to 0.48 unseen; seed range at most 0.024). Told $C' = C$, no design returns its input (0.19 to 0.22, bar 0.1).

The depth law.

The real arms obey it within 0.71% (84 pairs). On unseen arm-to-arm pairs the direct map misses it (median $|\log_2|$ scale error 0.195, seed range 0.061); two trained maps chained through the base come closer (0.125, seed range 0.022).

Arm-to-arm training (complete).

One added arm-to-arm pair per track (ratio 4) brings the held-out ratio-2 error to 0.148 (seed range 0.123), closing 0.674 of the gap to the

chained map (bar 0.5); it passes the transfer bar (0.148 against 0.367) and keeps the base fit (0.063 against 0.052). But the predicted distribution gets worse: CRPS 0.573 (0.466 + 0.107, seed range 0.062) against 0.525 (0.473 + 0.052) direct and 0.435 (0.429 + 0.006) chained, a gap closed of −0.53, mostly scale error in DNase. Both arms are complete and both controls pass, so the reading is final: fail (not judged). The every-pair second arm (not checked) reaches the chained map: CRPS 0.428 (0.425 + 0.003, seed range 0.005), gap closed 1.08.

The floor, and where settings act.

Whole-read thinning reproduces the real depth halves (1 kb shape ratio 1.002, against 1.226 for count thinning). At 3.75M reads per half the narrow shape floor is 0.79, not 0.69 (Fig. 2b; seed range at most 0.003). Signal level explains 0.18 to 0.38 of the per-bin disagreement; regional features add at most 0.0003 (shuffle 0.00003). Depth acts uniformly: its residual equals its whole-read null (DNase 0.774 against 0.773). The mapping filter does not: feature gain 0.099 to 0.110 (Fig. 2c), against at most 0.00009 for the floor.

4 Conclusion

On trained pairs the covariates carry the information: for depth they close most of the gap to the oracle; for run type and control settings they beat a covariate-free map. On unseen pairs the gain shrinks and the direct map misses the depth law. Two arm-to-arm pairs fix the law but not the distribution; every pair fixes both (not checked). Table 1 gives each goal's status. Limitations: 7 tracks, 3 seeds, 2 test chromosomes, and no ceiling for the run-type and control claims.

What carries to CANDI.

Use the full-depth track as the depth base, with depth as a fixed offset and no positional term. An NB-like, level-dependent spread describes the resampling floor. Read length and mapping filters act in low-mappability regions, so their correction needs sequence or mappability input. Conditioning will not carry to setting combinations that training never paired.

Next steps, without ranking:

- **A full-depth oracle for run type and control settings.** Moderate; gives those claims a ceiling.
- **Augment CANDI with counterfactual pairs.** Costly; tests whether the trained-pair gain carries into CANDI.

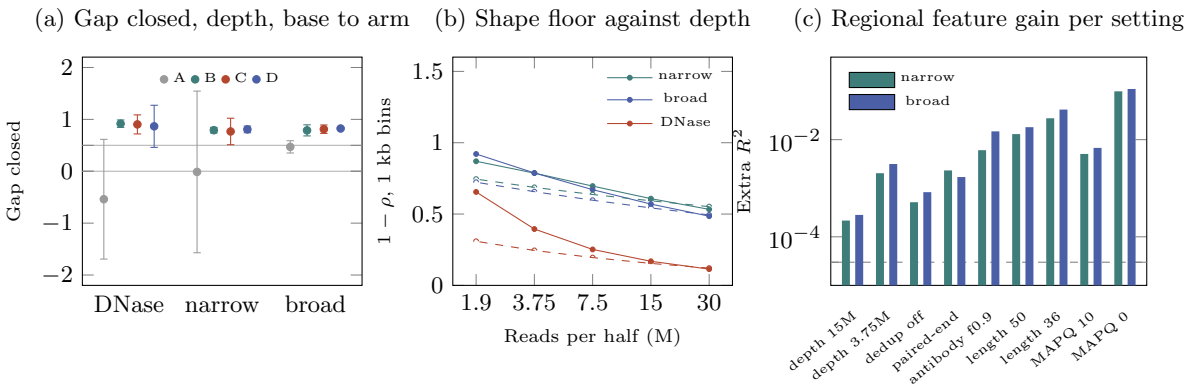


Fig. 2 (a) Gap closed from one-warp to the nested-read oracle on depth pairs, base to arm, one generator per track. Error bars are 2× the seed range in gap units; grid lines mark zero and the 0.5 bar. (b) The 1 kb shape floor (median over products) along the thinning ladder; the rightmost point is extrapolated. Solid lines thin whole reads, dashed lines thin single counts; the seed range is at most 0.003, smaller than the marks. (c) Extra R^2 that GC, mappability, repeats and peak context add to the signal-level null on the one-warp residual (median over tracks); the dashed line is the largest shuffle control.

Table 1 Each original goal, its status and one line of evidence. Status is a proposal for the PI to sign. Met: the question has a full reading. Partly met: part of it is read. Answered: the aim was not met, but the reading answers the question well enough to act.

Goal	Status	Evidence (not judged)
Do covariates map one measurement onto another? (main question)	Partly met	Beat the scrambled twin on trained pairs; count gain 0.29 falls to 0.11 unseen; swap and depth-law checks unmet
Depth conditioning beyond one-warp	Met	Designs B–D close 0.77–0.92 of the gap; gain above 2× seed range in 18/18 cells; design A no gain
Run-type conditioning beyond one-warp	Partly met	Beats one-warp in 16/16 histone cells by 0.055–0.213; no valid ceiling; DNase 1/8
Read-selection settings	Partly met	Count gain over twin 0.08–0.18 trained (seed range 0.003–0.059); p gain −0.26–0.01 unseen; one-warp bar not scored
Control and caller settings	Partly met	p gain 0.36–0.62 trained, 0.31–0.48 unseen (seed range $\leq$ 0.024); oracle interval negative, so bar unreadable
Capacity of the map	Answered	A monotone curve suffices in counts; more capacity does not fix unseen pairs; p explodes in B–D
Depth offset	Answered	Helps design A only: DNase CRPS 3.06 (1.36 + 1.70) to 1.247 (1.109 + 0.138), gain 1.813 against bar 0.227; other designs 0/36; depth law met in 0/64 cells
Counts-to-p map	Answered	Caller's formula plus learned correction, log CRPS 0.0659 ± 0.0035 (0.0656 + 0.0003), against 0.1791 (0.1367 + 0.0423) for an affine map; one track
Why the depth law is missed	Partly met	Not the pseudocount, loss or model size; arm-to-arm training meets the law (gap closed 0.674) but CRPS fails (−0.53); grid complete, controls pass; every-pair arm closes 1.08 (not checked)
Does CANDI's training let it correct other covariates?	Partly met	Told read length or run type, CANDI moves its level; per-region tracking unreadable (floor failed its depth control); probe parked
Which settings move a track most	Met	On one shared peak set: peak height above floor 27/27 (33–763×), shape 3/27
How much resampling alone moves a track	Met	1 kb shape floor at 30M: narrow 0.533, broad 0.485, DNase 0.114 (seed range $\leq$ 0.003); level, not region, sets it
Where along the genome settings act	Met	Depth uniform (residual = null); mapping filter 0.099–0.110, read length 0.013–0.042 extra R^2, in low-mappability bins